Chapter 32

TPN Primer for the Pharmacist: Approach to Nutrition Therapy

Mark Decerbo[1,2]
[1]Roseman University of Health Sciences, Henderson, NV, United States;
[2]University of Nevada School of Medicine, Reno, NV, United States

Learning Objectives:

Objective 32.1 Explain the need of parenteral nutrition.
Objective 32.2 Plan administration, calculation, and preparation of parenteral nutrition.

OBJECTIVE 32.1. EXPLAIN THE NEED OF PARENTERAL NUTRITION

Parenteral nutrition (PN) is a critical therapeutic modality used for a number of indications in neonates, children, and adults in which the pharmacist can play a major role in ensuring appropriate patient outcomes. PN is a complex medication containing as many as 40 separate components and is associated with numerous medication errors as well as significant adverse events, including death, and is therefore classified as a high-risk medication by the Institute for Safe Medication Practices (ISMP). Because of this, it is essential that the practicing pharmacist has a deep understanding of the provision of PN support, including appropriate indications for PN, selection of an appropriate formula, steps for safe initiation, approach to daily monitoring, need for modification to the prescribed PN formula, and awareness of potential complications. To achieve this, pharmacists are required to possess an advanced knowledge of pathophysiology, volume status, protein and energy requirements, macronutrients, and micronutrients to optimize the appropriate use of PN across all patient populations.[1,2]

Determine if Parenteral Nutrition Needed

The centerpiece of nutrition support is to use the least invasive and most appropriate method of feeding, which may not always be PN. The provision of macronutrients (e.g., protein/amino acids, carbohydrate, fat) and electrolytes directly via the intravenous route is fraught with potential issues, not the least of which are risks associated with central venous catheter placement. Because of this, PN is the overall the least preferred method of providing nutrition support, with enteral nutrition generally preferred. However, for appropriate patients, PN is the only viable means to provide nutrition and can therefore be seen as a potential life-saving measure. Due to the increased risk of complications with PN therapy, the pharmacist should first carefully assesses whether PN is appropriate for a given patient prior to proceeding with preparation of the formula.

General contraindications for PN include a functioning gastrointestinal (GI) tract, an inability to obtain venous access, a terminal prognosis which does not warrant aggressive care, and anticipated treatment for less than 5 days in a patient without severe malnutrition. Conversely, PN is generally indicated with the inability to absorb nutrients from the GI tract (which can occur in such conditions as short bowel syndrome, massive small-bowel resection, severe diarrhea, radiation enteritis, and untreatable malabsorption), complete bowel obstruction, intestinal pseudoobstruction, inability to obtain enteral access for enteral feeding, GI hemorrhage, high output enterocutaneous fistula where enteral access distally cannot be established, severe catabolism with expected duration of nill per oral (NPO) status $\times 5-7$ days, and pancreatitis where jejunal feeds cause intolerable abdominal pain.[3,4]

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Assessing the Patient

An assessment of malnutrition is a multifactorial process encompassing weight change, recent oral intake, functional capacity, symptoms, as well as a physical exam looking for presence of fat loss, muscle wasting, and edema. Recognition of malnutrition is key, as it is associated with an increased incidence of nosocomial infections, a higher rate of surgical complications, higher hospital costs, increased mortality, and increased length of stay. Prior to initiating PN, a nutrition assessment is therefore necessary to determine nutrient needs due to the patient's underlying condition(s). Factors that predispose to weight loss and malnutrition must be identified and generally fall into one of four categories: (1) Medical illnesses (sepsis, chronic disease, recent surgery/trauma), (2) Psychosocial factors (poverty, alcoholism, drug abuse, disability), (3) GI disorders (anorexia, dysphagia, persistent nausea/vomiting/diarrhea, inflammatory bowel disease, pancreatitis, GI fistulas), and (4) Medications (chemotherapy, appetite suppressants).[5,6]

Generally, a patient is considered malnourished if there is an established lack of enteral intake $\times 7-10$ days, 5% weight loss over 1 month, or 10% weight loss over 6 months. Scoring tools, such as the Subjective Global Assessment, Mini-Nutritional Assessment, and Prognostic Inflammatory and Nutritional Index, have been developed to predict nutritional status. More commonly, however, are laboratory assessments which can also be used to investigate the presence of malnutrition through an examination of acute phase proteins. These hepatic proteins can be useful nutritional markers but are not foolproof and may not truly reflect current nutritional status. The most commonly used markers are transferrin, albumin, and prealbumin, all of whom have differing half-lives (albumin 18−22 d, prealbumin 2−4 d, transferrin 7−10 d), and all of whom which can be falsely increased or decreased by associated factors. Volume depletion and renal failure can falsely elevate, whereas volume overload and iron deficiency, liver disease, pregnancy, cancer, trauma, and alcohol abuse can falsely suppress via general inflammation leading to an increase in TNF and Eicosanoids, which in turn leads to capillary leak and resultant low hepatic proteins. Despite these limitations, serum albumin level remains an important marker with studies showing progressively lower serum albumin correlated with increased ICU stay, increased hospital days, and even increased mortality. Approach to nutritional assessment in hospitalized patients with obesity is mentioned in the guidelines.[7]

A patient's current weight and weight history is therefore critical to consider when first addressing a patient's nutritional status. This information will assist the pharmacist in establishing the degree of malnutrition which may exist, and such can be measured against a variety of standards, including Ideal Body Weight, Ideal Creatinine Excretion, Upper Arm Anthropometry, and Body Mass Index. In addition to assessing degree of malnutrition, the patient's true weight will also form the basis by which caloric needs and protein requirements are calculated. If a falsely high weight is used, whether from operator error, patient history, or other variables such as volume status, such can lead to the overprovision of calories and the attendant risks and metabolic complications associated with overfeeding. The opposite is also true, with a falsely low reported weight leading to underfeeding, which can contribute to metabolic stress and morbidity and mortality in severely malnourished patients.[8]

Determining calorie and protein needs in malnourished patients is difficult, as matching caloric expenditure to caloric provision can be imprecise. Calorie requirements often increase in relation to stress, trauma, burns, and fever, whereas a decrease in needs may be seen in sedated and ventilated patients. Providing either more or fewer calories than that which is required by the patient can lead to poor outcomes. Overall, however, when determining the appropriate needed calorie level in patients, providing lower calorie levels initially appears to decrease risks associated with overfeeding, so long as an acceptable minimal level of calories is supplied. Indirect calorimetry is considered the gold standard when determining a patient's caloric expenditure; however, a wide variety of formulas and calculations are frequently used in practice to estimate the true amount of calories and protein which should be provided to the patient via their PN treatment. These predictive equations for energy needs each come with advantages and disadvantages, with the most commonly used equations being the: Harris−Benedict Equation, Ireton Jones Equation, Penn State Equation, and Mifflin St. Jeor Equation.[9−11] Regardless of which calculation is used by the pharmacist to estimate nutrient requirements, there is a notably convincing lack of evidence linking a given calorie and protein amount to patient outcomes, and the debate among healthcare professionals over the "best" equation to use continues in modern day practice. In general, most acutely ill patients require between 25 and 30 kcal/kg/day, with most critical care patients targeted to receive around 25 kcal/kg/day. Notable exceptions to this rule of thumb include obese patients (defined as $\geq$130% of their IBW), who are generally limited to 15−20 kcal/kg/day of their adjusted body weight, and patients at risk for refeeding syndrome (RFS) who are targeted to receive 15−25 kcal/kg/day.[12]

OBJECTIVE 32.2. PLAN ADMINISTRATION, CALCULATION, AND PREPARATION OF PARENTERAL NUTRITION

Once a decision has been made to appropriately proceed with PN, and a total goal number of calories have been calculated, the pharmacist must next verify the type of line/venous access which is present in the patient, either peripheral or central. Central PN is defined as nutrients provided intravenously which are delivered into a high flow vein, usually the superior vena cava adjacent to the right atrium. This is most commonly achieved through placement of a peripherally inserted central catheter line in the patient, although several other types of central access, each with their own advantages and disadvantages, are available. Conversely, peripheral PN, commonly referred to as PPN, is defined as nutrients provided intravenously which are delivered into a peripheral vein, usually of the hand or forearm via a peripheral line (short or midline).

PPN is utilized when central venous access is not possible and/or desired but is fraught with inherent nutritional limitations. It is optimally used only in select patients, namely those who require short-term (less than 10−14 days) nutrition support due to patency issues with this type of access. Additionally, patients being considered for PPN must generally possess adequate cardiac and renal function, as high volumes of PN must often be used to meet the patient's macronutrient goals, as the final protein concentration of the admixture must be <4.25%, whereas the final dextrose must be <10%. Lastly, due to risk of thrombophlebitis and the risk for loss of venous access due to hypertonicity, calculated osmolarity of the PPN formulation must never exceed an osmolarity of 900 mOsm/L, generally requiring <1800 kcal to be delivered via the formula, which may limit the appropriateness of this type of therapy. PPN can also be used adjunctively to diet in a transition period where a patient's oral intake is improving but is still suboptimally meeting the desired macronutrient balance.[13]

When administering PPN, the patient's peripheral line should be changed every 2−3 days, so as to minimize loss of patency and/or infection risk. The pharmacist can recommend several interventions in an attempt to maximize patency of the line, including addition of hydrocortisone and/or heparin the PN formula, as well as application of a transdermal nitroglycerin patch placed close to the catheter site. Drug−nutrient interactions should be avoided or monitored.[14] Even with all the aforementioned interventions, peripheral line patency may not be maintained, thus necessitating the potential need for central PN. Of note, there are several premixed commercially available fixed concentration PPN products containing dextrose and amino acids in varying strengths which are appropriate for peripheral use. Available with or without electrolytes, these products can play an important role in facilities and institutions in which the technical ability to make custom PN products is limited, as such ensure the potential for some, albeit limited, ability to provide some measure of nutritional support.

Central PN is provided via a venous catheter to large diameter veins with high rates of blood flow. Because of this, central PN allows infusion of formulations with osmolarities greater than 900 mOsm/L and are the route of choice for patients with cardiac or renal impairment, patients with poor peripheral venous access, and those who require PN for >10−14 days.[13]

Once the pharmacist has confirmed that the appropriate vascular access is present, they must next confirm and/or calculate the appropriate dose of each macronutrient for the individual patient, considering all current diagnoses, existing other therapies, and the targeted number of kcals/day. The doses of macronutrients used in PN should be mainly targeted at supporting protein synthetic response, while also addressing caloric support and metabolic management. Inappropriate provision of nutrients to a given patient can be dangerous, a fact which is complicated by the body's response to starvation and stress. In these conditions, metabolic responses are altered. Stressors can cause a breakdown in protein stores along with an accompanied increase in fat oxidation and energy expenditure. By accelerating loss of protein stores, patients are at higher risk for malnutrition. When designing the appropriate PN formula for a PN patient, the pharmacist must therefore carefully consider appropriate amounts of protein, carbohydrate, and fat by which to provide via the admixture. The process of calculating a patient's needed protein and caloric requirement is fairly straightforward, based primarily on a correct patient weight, along with analysis of the patient's degree of ongoing metabolic stress and body's ability to process macronutrients.[8] When ordering the PN solution, the following macronutrients should always be ordered as "amount per day" expressed as grams per day. Other methods, such as "amount per kg" or "% per PN bag," have let to sometimes catastrophic errors and should be avoided.

Protein

Amino acids, commercially supplied in numerous formulations (standard, hepatic failure, pediatric, renal) of varying percentages, provide 4 kcal/g. With acute metabolic stress, there is a significant increase in the amount of protein breakdown, thereby resulting in an increased amount of urea production. Patients requiring PN generally therefore have increased requirements, with protein/amino acids typically supplied in a range depending on the degree of stress and amount of catabolism which may be present. These ranges differ between adult and pediatric patients and are calculated on a gram per kilogram of euvolemic body weight basis. Neonates generally require between 2.5 and 3.0 g/kg/day, infants require 2.0−3.5 g/kg/day, and children require 1.5−2.0 g/kg/day. In adults, critical care, burn, surgery, and dialysis patients usually require protein administration in the range of 1.2−2 g/kg/day, extending up to 2−2.5 g/kg/day in severely catabolic states. Acute renal failure can require protein supplementation in the range of 1.5−1.6 g/kg/day due to protein losses through the glomeruli or hemodialysis; however, in renal failure patients who are not being dialyzed, protein restriction to as low as 0.8 g/kg may be needed to avoid elevating BUN to such a degree that complications arise.[15] Optimal degree of protein administration in patient with liver disease is somewhat controversial. Although dietary protein is converted by colonic bacteria into ammonia, intravenous amino acids in PN will not increase ammonia generation in the colon. Therefore, in patients with hepatic impairment, protein goals are generally suggested to be between 1.0 and 1.5 g/kg, with a reduction to 0.8 g/kg only if hepatic encephalopathy resistant to medication management is present.[16]

Carbohydrate

Dextrose, providing 3.4 kcal/g and commercially supplied in solutions of varying percentages, is the only carbohydrate utilized in PN. In stressed patients, there is an increased need for glucose accompanied by impaired insulin secretion and sensitivity, leading to the potential for hyperglycemia. Therefore, in most PN formulations which are infused over 24 h, carbohydrates are generally provided in varying amounts up to a maximum of 60% of total kcals/day. Initial dextrose in these PN solutions should generally not exceed 7.2 g/kg/day (5 mg/kg/min). This parameter is known as the glucose infusion rate (GIR) and is used as a benchmark to minimize the risks of overfeeding with dextrose, which are primarily the occurrence of fatty liver and hyperglycemia. In circumstances where patients are receiving a cyclic PN solution, that which is infused over less than 24 h per day, patients may exceed the above GIR when additional calories for weight maintenance or gain are desired. Conversely, due to the risk associated with hyperglycemia in the critically ill patient, dextrose in this population is generally limited to no more than 5 mg/kg/day. In patients with history of diabetes, insulin therapy should be initiated for blood glucose more than 140 mg/dL. In patients without diabetes, insulin therapy is indicated if blood glucose is above 180 mg/dL or in patients with persistent requirement for correction insulin.[17]

Fat

Injectable lipid emulsions (ILE), formerly known as intravenous lipid emulsions (IVFE), arise from various oil sources, including soybean oil, olive oil, fish oil, safflower oil, and medium chain triglycerides. Irrespective of the selected commercially available product, lipids should generally provide 20%−30% of total daily kcals or <2.5 g/kg/day in acute care patients, whereas critical care patients are generally limited to 0.4−0.75 g/kg/day. Exceptions exist, the most notable being precluding concomitant conditions, such as concurrent treatment with propofol and/or clevidipine infusions. These medications can decrease a patient's need for supplemental lipid in the PN solution, as the vehicle for both medications provides essential fatty acids and calories. Additionally, hypertriglyceridemia, with the risk for acute pancreatitis when markedly elevated, can also serve to limit the amount of lipid provided to the patient as well. In all cases, however, if a patient is to be on PN for greater than 3 weeks, a minimum of 2%−4% of total calories should come from IVFE to prevent essential fatty acid deficiency. Of the various commercially available lipid products, naturally occurring Vitamin K varies depending on the manufacturer and concentration, with safflower oil containing less vitamin K than soybean oil. Vitamin K typically increases proportionally with increasing lipid concentrations and must be considered by the pharmacist when used in a patient on concomitant warfarin therapy.[18]

Micronutrients

Micronutrients are generally defined as essential nutrients (vitamins, electrolytes, trace elements) required daily in small (milligram, microgram) amounts. Vitamins are a vital component of any PN admixture and should generally be supplemented based on daily recommended doses for adults and children. To achieve this, vitamins are usually commercially

supplied as standardized doses formulated to meet daily requirements and prevent toxicity. These prespecified formulations generally meet the needs of the majority of patients, with modifications possible for select situations. Of note, thiamine is perhaps the most important vitamin to consider, as it is not only one of the first vitamins to manifest with symptoms of acute deficiency but has been implicated as a cause of death in several patients receiving thiamine-deficient PN therapy. Of the deaths, each patient developed lactic acidosis and died within 5 weeks of initiation of treatment with thiamine-deficient TPN. Thiamine is essential for two enzymes needed for aerobic metabolism, and without sufficient amounts, pyruvate cannot enter the Krebs cycle, resulting in pyruvate accumulation and conversion into lactate. For these reasons, prevention and recognition of thiamine deficiency is critical.[19,20]

Trace minerals, similar to multivitamins, are commercially formulated in quantities expected to match patient needs. These are especially important in providing adequate PN formulations in long-term PN patients, as deficiencies can result when there is no oral intake and trace minerals are withheld from the PN formulation. Although infused in minute quantities, if not carefully monitored in these patients, they can have detrimental effects through accumulation and resultant toxicity. It is vital that the pharmacist understand dosages available, patients' conditional requirements, and the importance of increasing or decreasing supplemental amounts based on patient-specific factors. Perhaps the most commonly employed trace element is zinc, which can commonly manifest via delayed wound healing and impaired immunity. Although the standard PN dose of zinc is 2.5−5 mg/day, patient with high GI losses may require upward of 12−17 mg/L of fluid lost, providing a concrete example of the importance of patient-specific trace mineral dosing.[19,20]

Electrolytes

Electrolytes in PN formulations are added according to anticipated patient requirements, demonstrated laboratory deficiencies, metabolic response to medications, and recommended daily intakes. Although there are various published recommendations which provide typical ranges for parenteral electrolyte content, a patient's age and renal function are the two main determinants of the daily dosage of electrolytes. Electrolyte needs can also vary widely based on losses from wounds, surgical drains, fever, emesis, GI suctioning, and diarrhea and must therefore be replaced either in the PN solution itself or as separate electrolyte riders. Electrolytes must be ordered as the complete salt form rather than the individual ion, with ingredients ordered as "amount per day" in units of mEq, mmol, mcg, or mg.

Parenteral Nutrition Preparation

Appropriate PN preparation requires the pharmacist to collect and monitor a multitude of information. Prior to proceeding with the actual compounding of the formula, the pharmacist should undertake a review to ensure that the following standards are met. If not, additional information and clarifications may be required to be obtained prior to proceeding any further. Sterile compounding of PN preparations is shown in Fig. 32.1.



FIGURE 32.1 Sterile compounding of parenteral nutrition. *Courtesy of Baxter Healthcare Limited.*

Required elements of the PN allow for order review:

- Patient information
  - Name, identifiers, date of birth, age, allergies, height, dosing weight, and diagnoses
- Indication for PN
- Route and access device for PN
- Prescriber contact information
- Date and time PN order submitted
- Date and time PN order to be administered
- Volume, rate of infusion, and infusion schedule for PN
- Method of ILE delivery (separate infusion or total nutrient admixture [TNA])
- PN ingredients for adults listed as amounts/day
- PN ingredients for neonates/pediatrics listed as amounts/kg/day
- PN ingredients
  - A dose for each macronutrient
  - A dose for each electrolyte salt
  - A dose for each multivitamin
  - A dose for each individual vitamin
  - A dose for each trace element
  - A dose for each nonnutrient medication, if ordered

Following receipt of all ordered components of the PN, the pharmacist must ensure that all ingredients are compatible with each other, and that the PN preparation is expected to be stable from the time of compounding through the end of the infusion. From a formula standpoint, the pharmacist must determine if all ordered components of the PN solution can be physically supplied based on desired final volume of the solution. This can be especially difficult in volume restricted patients, at times impossible to deliver the desired macronutrients and additives based on stated fluid restrictions.[13]

From a stability standpoint, certain components in PN admixtures may be incompatible. This most often occurs in orders where highly concentrated PN admixtures are ordered. There is a well-known physiochemical incompatibility between calcium and phosphate salts, which can react to produce dangerous insoluble products known as precipitates. Minimizing the amount of calcium or phosphate can reduce the risk of precipitation, but this is not always practical based about patient-specific needs. Numerous other factors may affect calcium and phosphorus compatibility as well, including temperature, pH of the final PN solution, selection of salt used, and final concentration of macronutrient additives. For this reason, acetate salts can be used in PN solutions, but never carbonate salts, which are known for the propensity to insolubilize calcium. When supplying calcium, the gluconate salt must be used preferentially over chloride salts, as it is least dissociated into its free form when mixed in PN solutions. Because of the multitude of factors which affect solubility, calcium phosphate limits must be evaluated with every PN formulation. Given the multiple variables that impact calcium−phosphate solubility, the use of simple equations to calculate calcium−phosphate limits which do not consider all factors should be avoided. Practitioners may instead consult published graphs or use validated software programs for specific products which provide adequate guidance when determining if the specific PN formulation is at risk for precipitation.[13,21]

In terms of effect of macronutrient concentration, PN solutions are primarily ultimately admixed in one of two different manners, due to the inherent instability of lipid emulsion. Although dextrose and protein are always admixed in the same container with one another, IVFE may need to be administered separately or have its dose adjusted to be stable in a TNA. Also known as a 3-in-1, a TNA is a PN formulation containing ILE as well as the other components of PN (carbohydrate, amino acids, vitamins, minerals, trace elements, water, and other additives) in a single container, whereas a 2-in-1 contains carbohydrates and amino acids (along with the other aforementioned components) except for lipid which is given as a physically separate infusion. In these cases, to ensure stability when using a TNA, in the absence of any manufacturer data, TNA should have a final concentration of amino acid formulations $\geq$4%, dextrose $\geq$10%, and soybean oil−based lipid emulsion $\geq$2%. These stability parameters may vary depending on the amino acid formulation and specific IV lipid emulsion from other oil sources. Ideally, TNAs should remain stable for up to 30 h at room temperature (25°C) or for 9 days refrigerated (5°C) followed by 24 h at room temperature. In the absence of formulation-specific emulsion stability data, it is recommended that a 2-in-1 admixture be compounded and a lipid be administered as a separate infusion.[22]

Lastly, the PN formulation must be assessed for potential incompatibilities with nonnutrient medications which have been included in the order. Nonnutrient medications must be included in PN only when supported by pharmaceutical data describing physicochemical compatibility and stability of the additive medication, as well as clinical data confirming the

expected therapeutic actions of the medication. Adding medications without the above-proven data can lead to unnecessary patient risks and is generally only considered routinely safe for regular insulin and H2 receptor antagonists. Hyperglycemia is the most common complication of PN, and for this reason, regular human insulin is commonly added to PN to aid in regulation of blood glucose. Other insulin products, while commercially available, are not compatible with PN solutions, and should be avoided. For a patient requiring insulin, most practitioners target the addition of 1−2 units per 10 g dextrose to the PN formulation. As needed, this amount should be increased daily by adding 75%−100% of the prior day's sliding scale insulin to the subsequent PN formulation. The timing and frequency of blood glucose monitoring while on PN must be based on the patient's clinical status and performed in a manner appropriate for the PN infusion schedule (24 h infusion vs. cyclic PN).[14]

H2 receptor antagonists are additionally often added to the PN admixture in patients requiring GI stress ulcer prophylaxis, of which famotidine and ranitidine are most common. Both are stable in 3-in-1 admixtures of various formulations and may be added without hesitation. However, it is not uncommon for potentially incompatible medications to be ordered as part of the PN solution, the most common of which is intravenous iron. Often required for issues related to the underlying need for PN therapy, iron has been shown to destabilize lipid formulations and thus should be preferentially supplemented separately when using 3-in-1 solutions. Whenever an incompatible medication is ordered, infusion of the PN admixture will need to be interrupted so the incompatible medication can be infused. After the PN infusion is stopped, the line should be flushed with a fluid compatible with the medication such as saline or dextrose. The medication may then be infused, and after the medication has been administered, the line is flushed again with the compatible fluid before the PN infusion is resumed.[14]

Serious consequences associated with the use of PN are well-known and can generally be divided into either short-term or long-term risks. In the short term, primary PN-associated risks include those presented by the placement of a central line, infection, and RFS. First described following the Second World War, RFS is one of the most dramatic and life-threatening side effects described in PN patients. Occurring primarily in severely malnourished patients who receive aggressively high amounts of calorie and carbohydrates in their PN therapy, RFS results from hormonal and metabolic changes. These changes induce rapid onset hypophosphatemia, hypomagnesemia, and/or hypokalemia which can lead to cardiac and respiratory failure and death. Strategies to reduce the risk of developing RFS include appropriate screening to identify at risk patients, aggressive prophylactic thiamine repletion, aggressive electrolyte repletion, as well as a gradual introduction of nonprotein kcals when initiating PN in severely malnourished patients, with patients titrated to caloric goal over a minimum 5−7 days.[23,24]

In the long-term, complications have been commonly reported with the excessive provision of macronutrients and/or total calories over time. These primary long-term complications of PN include fatty liver, cholestasis, metabolic bone disease, and electrolyte/vitamin/mineral deficiency or toxicity. In the long-term PN patient, it is important to be aware of, monitor for, and attempt to address these adverse effects. Approximately 1 in 5 deaths in patients on long-term PN is due to liver failure, and excess caloric provision from either dextrose or lipid sources is thought to be the cause. With dextrose overfeeding, the excess dextrose causes hyperinsulinemia, which in turn enhances glucose conversion into fat within the liver. Because of this, recommendations for decreasing rates of TPN included liver disease include limiting dextrose to 4 g/kg/day. Although essential fatty acid deficiencies contribute to the development of hepatic complications in patients on long-term PN, it is equally important to avoid overfeeding with lipids, generally limiting such to 30% of total daily calories through supplementation of no more than 1 g/kg/day. At signs of liver dysfunction, lipids can be further reduced while ensuring no deficiencies result. Lastly, supplementation with carnitine, taurine, glutamate, and choline may also help to decrease the long-term liver effects seen with PN, as can a trial of cyclic PN, in which PN therapy is infused over periods of 8−16 h per day, rather than the standard 24 h.[25]

PN should be discontinued once the patient has been transitioned to either PO or enteral nutrition, although in some patients, such as those with short-gut syndrome, therapy will be lifelong. Generally, once oral intake has advanced to approximately 50% of estimated kcals needs, the PN formula can be weaned or discontinued. Attention to glycemic control is critical in those patients who were treated with insulin as part of their PN formulation and/or those who developed new onset hyperglycemia while on PN therapy. In clinical practice, when discontinuing PN, formulations are routinely tapered off to prevent rebound hypoglycemia. In adults, such a process is most likely unnecessary, especially if the patient is receiving carbohydrates through their oral diet. Nevertheless, this tapering can be practically accomplished by cutting the infusion rate of the PN solution by 50% over the final 1−2 h of infusion prior to turning off completely. If the infusion of an ILE being given separately from the 2-in-1 infusion needs to be discontinued, no rate reduction prior to turning off is necessary.

Prior to dispensing the final completed PN product to the patient or third-party service, the pharmacist should undertake a series of steps to ensure that both safety and quality measures are in place. The original PN order with patient-specific

label and compounding formulation label should again be verified, with all patient-specific and auxiliary labels affixed to the PN container. The pharmacist should next visually inspect the PN container and contents for evidence of precipitation or lipid destabilization (cracking, creaming, oiling out of the emulsion) or physical leaks to the bag, with the PN formula remade if any of the aforementioned issues are observed. Next, the pharmacist should place the prepared PN admixture under the appropriate storage conditions (refrigeration, out of light) while on the way to delivery.[13]

CONCLUSION

PN can be a life-saving modality, but as with all medical interventions, diligent monitoring and correct administration are required to be successful. PN is a valuable and necessary medical treatment for many patients, but when used suboptimally through knowledge deficits or oversight can present a host of risks to the patient. By focusing on the core essentials of PN management from the pharmacist's perspective, this form of nutrition support can be applied successfully to optimize patient outcomes.